EE 467 Lab 2: Breaking CAPTCHAs with Tensorflow / Keras

In lab 2, we are going to work on automatically breaking CAPTCHAs with deep learning! Originally, CAPTCHA stands for "Completely Automated Public Turing test to tell Computers and Humans Apart", and traditional CAPCTHAs serve as a great tool to stop spam bots and malicious crawlers. Today, as we have made huge progress in computer vision and deep learning, these CAPCTHAs are no longer unbreakable by computers. Now let's prove the correctness of the claim by ourselves!

As usual, please check if the helper library, `lab_2_helpers.py` and the extracted dataset directory, `captcha-images` exist under the same directory. Then, please install all libraries used for this lab:

```
In [1]: %pip install matplotlib scikit-learn "opencv-python>4" imutils

        # Install Tensorflow with CPU support only, or ...
        %pip install "tensorflow>2"
        # Install Tensorflow with GPU support:
        #%pip install tensorflow-gpu>2
```

```
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: opencv-python>4 in /usr/local/lib/python3.12/dist-packages (4.12.0.88)
Requirement already satisfied: imutils in /usr/local/lib/python3.12/dist-packages (0.5.4)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: numpy>=1.23 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.1)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Requirement already satisfied: tensorflow>2 in /usr/local/lib/python3.12/dist-packages (2.19.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (1.4.0)
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (0.7.0)
Requirement already satisfied: google-pasta>=0.1.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (18.1.1)
Requirement already satisfied: opt-einsum>=2.3.2 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (3.4.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (25.0)
Requirement already satisfied: protobuf!=4.21.0,!=4.21.1,!=4.21.2,!=4.21.3,!=4.21.4,!=4.21.5,<6.0.0dev,>=3.20.3 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (5.29.5)
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (2.32.4)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (75.2.0)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (3.3.0)
Requirement already satisfied: typing-extensions>=3.6.6 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (4.15.0)
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (2.0.1)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (1.76.0)
Requirement already satisfied: tensorboard~=2.19.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (2.19.0)
Requirement already satisfied: keras>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (3.10.0)
Requirement already satisfied: numpy<2.2.0,>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (2.0.2)
Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (3.15.1)
Requirement already satisfied: ml-dtypes<1.0.0,>=0.5.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow>2) (0.5.4)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from astunparse>=1.6.0->tensorflow>2) (0.45.1)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from keras>=3.5.0->tensorflow>2) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.12/dist-packages (from keras>=3.5.0->tensorflow>2) (0.1.0)
Requirement already satisfied: optree in /usr/local/lib/python3.12/dist-packages (from keras>=3.5.0->tensorflow>2) (0.18.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow>2) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow>2) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow>2) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow>2) (2026.1.4)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.12/dist-packages (from tensorboard~=2.19.0->tensorflow>2) (3.10)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from tensorboard~=2.19.0->tensorflow>2) (0.7.2)
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from tensorboard~=2.19.0->tensorflow>2) (3.1.5)
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.12/dist-packages (from werkzeug>=1.0.1->tensorboard~=2.19.0->tensorflow>2) (3.0.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich->keras>=3.5.0->tensorflow>2) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich->keras>=3.5.0->tensorflow>2) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich->keras>=3.5.0->tensorflow>2) (0.1.2)
```

Next, we import all tools needed before starting:

```
In [34]: import os, pickle, glob, math
         from pprint import pprint

         import cv2
         import numpy as np
         import imutils
         from imutils import paths
         from tensorflow.keras import Sequential, layers
         from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense
         from matplotlib import pyplot as plt
         from matplotlib.gridspec import GridSpec
         from sklearn.preprocessing import LabelBinarizer
         from sklearn.model_selection import train_test_split

         from lab_2_helpers import *
```

Preprocessing

Ground Truth Characters Extraction

As usual, we will start pre-processing stage by loading CAPTCHA images into the memory:

```
In [4]: !tar -xJf captcha-images.tar.xz
```

```
In [5]: # Dataset images folder
        CAPTCHA_IMAGE_FOLDER = "./captcha-images"

        # List of all the captcha images we need to process
        captcha_image_paths = list(paths.list_images(CAPTCHA_IMAGE_FOLDER))
        # Review image paths
        pprint(captcha_image_paths[:10])
```

```
['./captcha-images/3F2L.png',
 './captcha-images/J3Q3.png',
 './captcha-images/2HKY.png',
 './captcha-images/3G9T.png',
 './captcha-images/B6RD.png',
 './captcha-images/6RTM.png',
 './captcha-images/2E8X.png',
 './captcha-images/2SBQ.png',
 './captcha-images/2NNC.png',
 './captcha-images/4NGH.png']
```

Note that for each image, its file name (without extension) happens to be its corresponding CAPTCHA text. Thus, we extract file names for all CAPTCHA images and save them as labels for future use:

```
In [6]: def extract_captcha_text(image_path):
            """ Extract correct CAPTCHA texts from file name of images. """
            # Extract file name of image from its path
            # e.g. "./captcha-images/2A2X.png" -> "2A2X.png"
            image_file_name = os.path.basename(image_path)
            # Extract base name of image, omitting file extension
            # e.g. "2A2X.png" -> "2A2X"
            return os.path.splitext(image_file_name)[0]

        captcha_texts = [extract_captcha_text(image_path) for image_path in captcha_image_paths]
        # Review extraction results
        pprint(captcha_texts[:10])
```

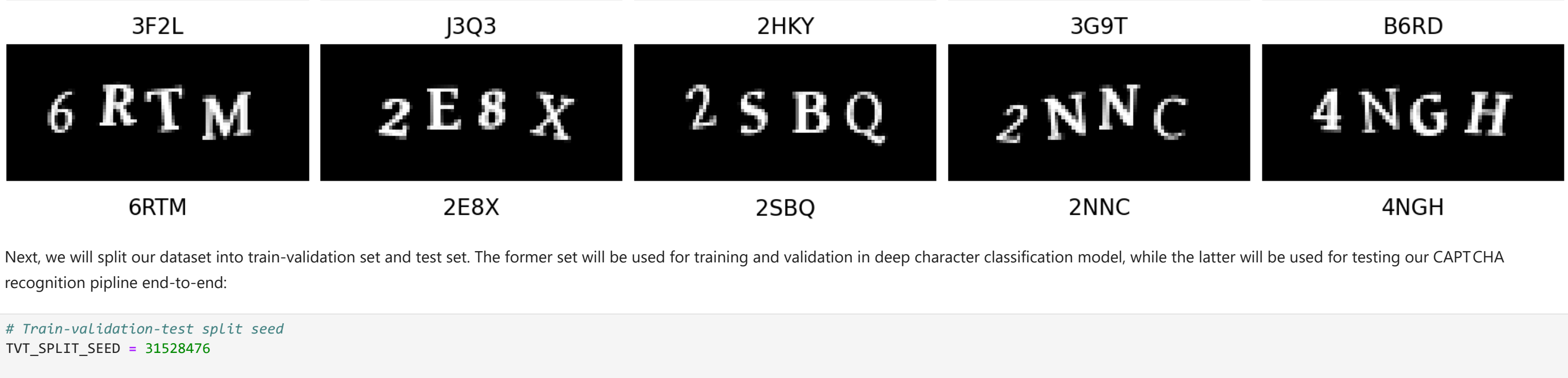
```
['3F2L', 'J3Q3', '2HKY', '3G9T', 'B6RD', '6RTM', '2E8X', '2SBQ', '2NNC', '4NGH']
```

Loading and Transforming Images

For the feature extraction stage, we are going to extract individual characters from these CAPTCHAs. This is done by looking for contours (bounding boxes) around characters, then cropping the CAPTCHAs such as only the contour areas are preserved. We begin feature extraction by loading and transforming images:

```
In [25]: import cv2

         def load_transform_image(image_path):
             """ Load and transform image into grayscale. """
             ## [ TODO ]
             # 1) Load image with OpenCV
             image = cv2.imread(image_path)

             # 2) Convert image to grayscale
             image_gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
             # 3) Add extra padding (8px) around the image
             image_padded = cv2.copyMakeBorder(image_gray,8, 8,8,8, cv2.BORDER_CONSTANT, value=(255,255,255))

             return image_padded

         captcha_images = [load_transform_image(image_path) for image_path in captcha_image_paths]
         # Review loaded CAPTCHAs
         print_images(
             captcha_images[:10], n_rows=2, texts=captcha_texts[:10]
         )
```

```
Figure(2000x500)
```


3F2L J3Q3 2HKY 3G9T B6RD

6RTM 2E8X 2SBQ 2NNC 4NGH

Next, we will split our dataset into train-validation set and test set. The former set will be used for training and validation in deep character classification model, while the latter will be used for testing our CAPTCHA recognition pipline end-to-end:

```
In [28]: # Train-validation-test split seed
         TVT_SPLIT_SEED = 31528476

         # Perform split on CAPTCHA images as well as labels
         captcha_images_tv, captcha_images_test, captcha_texts_tv, captcha_texts_test = train_test_split(
             captcha_images, captcha_texts, test_size=0.2, random_state=TVT_SPLIT_SEED
         )

         print("Train-validation:", len(captcha_texts_tv))
         print("Test:", len(captcha_texts_test))
```

```
Train-validation: 908
Test: 228
```

Bounding Box Extraction

It's now time to perform the most important feature extraction step: finding contours and extracting characters. Contours can be explained simply as a curve joining all the continuous points (along the boundary), having same color or intensity. It is useful for shape analysis and object detection and recognition. For our task however, we are **more interested in the bounding boxes around characters**, since these are the part of images we will be used for character classification:

Here are the steps we are going to perform:

- Find contours around characters in CAPTCHAs.
- Get the position and size for each corresponding bounding box.
- If a bounding box is too wide (width-to-height ratio larger than 1.25), chances are that we have bounded two letters in a single bounding box. In this case, split the bounding box vertically from the center into two.
- Store all bounding boxes for the CAPTCHA.
- If there aren't 4 bounding boxes for the CAPTCHA, ignore it since our character extraction process must have run into problems in this case.
- Sort the bounding boxes by their X coordinates, so that they match the order corresponding letters occur.
- For each bounding box, extract corresponding region of the image, and store it as an instance of corresponding character at `${CHAR_IMAGE_FOLDER}/{letter}/{count}.png`.

After these steps, we have transformed CAPTCHA images into images of single character. This simplifies our task since now our model only needs to deal with classification (from character image to character itself) rather than also dealing with detection (finding and extracting charatcers).

```
In [29]: # Character images folder template
         CHAR_IMAGE_FOLDER = f"./char-images-{TVT_SPLIT_SEED}"

         def extract_chars(image):
             """ Find contours and extract characters inside each CAPTCHA. """
             # Threshold image and convert it to black-white
             image_bw = cv2.threshold(image, 0, 255, cv2.THRESH_BINARY_INV | cv2.THRESH_OTSU)[1]
             # Find contours (continuous blobs of pixels) the image
             contours = cv2.findContours(image_bw, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)[0]

             char_regions = []
             # Loop through each contour
             for contour in contours:
                 # Get the rectangle that contains the contour
                 x, y, w, h = cv2.boundingRect(contour)

                 # Compare the width and height of the bounding box,
                 # detect if there are letters conjoined into one chunk
                 if w / h > 1.25:
                     # Bounding box is too wide for a single character
                     # Split it in half into two letter regions
                     half_width = int(w / 2)
                     char_regions.append((x, y, half_width, h))
                     char_regions.append((x + half_width, y, half_width, h))
                 else:
                     # Only a single letter in contour
                     char_regions.append((x, y, w, h))

             # Ignore image if less or more than 4 regions detected
             if len(char_regions)!=4:
                 return None
             # Sort regions by their X coordinates
             char_regions.sort(key=lambda x: x[0])

             # Character images
             char_images = []
             # Save each character as a single image
             for x, y, w, h in char_regions:
                 # Extract character from image with 2px margin
                 char_image = image[y - 2:y + h + 2, x - 2:x + w + 2]
                 # Save character images
                 char_images.append(char_image)

             # Return character images
             return char_images

         def save_chars(char_images, captcha_text, save_dir, char_counts):
             """ Save character images to directory. """
             for char_image, char in zip(char_images, captcha_text):
                 # Get the folder to save the image in
                 save_path = os.path.join(save_dir, char)
                 os.makedirs(save_path, exist_ok=True)

                 # Write letter image to file
                 char_count = char_counts.get(char, 1)
                 char_image_path = os.path.join(save_path, f"{char_count}.png")
                 cv2.imwrite(char_image_path, char_image)

                 # Update count
                 char_counts[char] = char_count+1

         # Force character extraction even if results are already available
         FORCE_EXTRACT_CHAR = False

         char_counts = {}
         # Extract and save images for characters
         if FORCE_EXTRACT_CHAR or not os.path.exists(CHAR_IMAGE_FOLDER):
             for captcha_image, captcha_text in zip(captcha_images_tv, captcha_texts_tv):
                 # Extract character images
                 char_images = extract_chars(captcha_image)
                 # Skip if extraction failed
                 if char_images is None:
                     continue
                 # Save character images
                 save_chars(char_images, captcha_text, CHAR_IMAGE_FOLDER, char_counts)
```

Label Encoding

During the training stage, we are going to load character images from previous stages as features and generate corresponding labels from their path. We will then rescale features, one-hot encode labels (occurred characters) and save labels to an external file.

```
In [30]: # Path of occurred characters (labels)
         LABELS_PATH = "./labels.pkl"

         def make_feature(image):
             """ Process character image and turn it into feature. """
             # Resize letter to 20*20
             image_resized = resize_to_fit(image, 20, 20)
             # Add extra dimension as the only channel
             feature = image_resized[..., None]

             return feature

         def make_feature_label(image_path):
             """ Load character image and make feature-label pair from image path. """
             # Load image and make feature
             feature = make_feature(cv2.imread(image_path, cv2.COLOR_BGR2GRAY))
             # Extract label based on the directory the image is in
             label = image_path.split(os.path.sep)[-2]

             return feature, label

         # Make features and labels from character image paths
         features_tv, labels_tv = unzip((
             make_feature_label(image_path) for image_path in paths.list_images(CHAR_IMAGE_FOLDER)
         ))

         # Scale raw pixel values into range [0, 1]
         features_tv = np.array(features_tv, dtype="float")/255
         # Convert labels into one-hot encodings
         lb = LabelBinarizer()
         labels_one_hot_tv = lb.fit_transform(labels_tv)
         # Number of classes
         n_classes = len(lb.classes_)

         # Further split the training data into training and validation set
         X_train, X_vali, y_train, y_vali = train_test_split(
             features_tv, labels_one_hot_tv, test_size=0.25, random_state=955996
         )
         # Save mapping from labels to one-hot encoding
         with open(LABELS_PATH, "wb") as f:
             pickle.dump(lb, f)
```

Training

Next, we build a Convolutional Neural Network (CNN) as our classification model with Tensorflow / Keras. The structure of the neural network is shown below:

After building the neural network, we train it and save weights.

```
In [36]: # Batch size
         BATCH_SIZE = 32
         # Number of epochs
         N_EPOCHS = 10

         # Path of model weights file
         MODEL_WEIGHTS_PATH = "./captcha-model.weights.h5"
         # Force training even if weights are already available
         FORCE_TRAINING = True

         # Build a feed-forward neural network
         model = Sequential()

         ## [ TODO ]
         # Implement the neural network shown above

         # First convolution block: (*, 20, 20, 1) -Conv2D+ReLU-> (*, 20, 20, 20) -MaxPooling2D-> (*, 10, 10, 20)
         # (Remember to include input shape for the first layer!)
         # 1) Convolution layer: 20 channels, 5*5 kernel, ReLU activation, padded to maintain same shape
         # 2) Max pooling layer: 2*2 kernel
         model.add(Conv2D(8, (5,5), strides=(1,1), padding="same",activation="relu",input_shape=(20,20,1)))
         model.add(MaxPooling2D(pool_size=(2,2), strides=(2,2)))

         # Second convolution block: (*, 10, 10, 20) -Conv2D+ReLU-> (*, 10, 10, 50) -MaxPooling2D-> (*, 5, 5, 50)
         # Convolution layer: same as above, but use 50 channels
         model.add(Conv2D(50, (5,5), strides=(1,1), padding="same", activation="relu", input_shape=(10,10,20)))
         model.add(MaxPooling2D(pool_size=(2,2), strides=(2,2)))

         # Flatten layer reshape features to 1D: (*, 5, 5, 50) -Flatten-> (*, 1250)
         model.add(Flatten())

         # First fully-connected (linear) layer: (*, 1250) -FC+ReLU -> (*, 500)
         model.add(Dense(500, activation="relu"))
         # Last fully-connected (linear) layer: (*, 500) -FC+Softmax-> (*, n_classes)
         # Softmax outputs a categorical distribution representing probability for each character
         model.add(Dense(32, activation="softmax"))

         # Build classification model with Adam optimizer and cross entropy loss
         model.compile(loss="categorical_crossentropy", optimizer="adam", metrics=["accuracy"])

         if FORCE_TRAINING or not os.path.exists(MODEL_WEIGHTS_PATH):
             # Train the neural network model
             model.fit(
                 X_train, y_train, validation_data=(X_vali, y_vali),
                 batch_size=BATCH_SIZE, epochs=N_EPOCHS, verbose=1
             )
             # Save model weights to disk
             model.save_weights(MODEL_WEIGHTS_PATH)
         else:
             model.load_weights(MODEL_WEIGHTS_PATH)

         # Show model summary
         print(model.summary())
```

```
Epoch 1/10
84/84 ━━━━━━━━━━━━━━━━━━━━ 5s 31ms/step - accuracy: 0.2103 - loss: 2.9422 - val_accuracy: 0.8642 - val_loss: 0.5260
Epoch 2/10
84/84 ━━━━━━━━━━━━━━━━━━━━ 2s 29ms/step - accuracy: 0.9162 - loss: 0.2928 - val_accuracy: 0.9641 - val_loss: 0.2222
Epoch 3/10
84/84 ━━━━━━━━━━━━━━━━━━━━ 3s 41ms/step - accuracy: 0.9856 - loss: 0.0929 - val_accuracy: 0.9753 - val_loss: 0.1627
Epoch 4/10
84/84 ━━━━━━━━━━━━━━━━━━━━ 3s 37ms/step - accuracy: 0.9890 - loss: 0.0526 - val_accuracy: 0.9764 - val_loss: 0.1284
Epoch 5/10
84/84 ━━━━━━━━━━━━━━━━━━━━ 4s 29ms/step - accuracy: 0.9924 - loss: 0.0333 - val_accuracy: 0.9820 - val_loss: 0.1267
Epoch 6/10
84/84 ━━━━━━━━━━━━━━━━━━━━ 2s 29ms/step - accuracy: 0.9975 - loss: 0.0155 - val_accuracy: 0.9832 - val_loss: 0.1193
Epoch 7/10
84/84 ━━━━━━━━━━━━━━━━━━━━ 5s 66ms/step - accuracy: 0.9965 - loss: 0.0176 - val_accuracy: 0.9854 - val_loss: 0.1119
Epoch 8/10
84/84 ━━━━━━━━━━━━━━━━━━━━ 3s 39ms/step - accuracy: 0.9979 - loss: 0.0085 - val_accuracy: 0.9854 - val_loss: 0.1100
Epoch 9/10
84/84 ━━━━━━━━━━━━━━━━━━━━ 4s 42ms/step - accuracy: 0.9997 - loss: 0.0030 - val_accuracy: 0.9843 - val_loss: 0.1213
Epoch 10/10
84/84 ━━━━━━━━━━━━━━━━━━━━ 5s 63ms/step - accuracy: 0.9991 - loss: 0.0061 - val_accuracy: 0.9865 - val_loss: 0.1121
Model: "sequential_3"
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━┓
┃ Layer (type)                    ┃ Output Shape           ┃       Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━┩
│ conv2d_2 (Conv2D)               │ (None, 20, 20, 8)      │           208 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ max_pooling2d_2 (MaxPooling2D)  │ (None, 10, 10, 8)      │             0 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ conv2d_3 (Conv2D)               │ (None, 10, 10, 50)     │        10,050 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ max_pooling2d_3 (MaxPooling2D)  │ (None, 5, 5, 50)       │             0 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ flatten_1 (Flatten)             │ (None, 1250)           │             0 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ dense_2 (Dense)                 │ (None, 500)            │       625,500 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ dense_3 (Dense)                 │ (None, 32)             │        16,032 │
└─────────────────────────────────┴────────────────────────┴───────────────┘
 Total params: 1,955,372 (7.46 MB)
 Trainable params: 651,790 (2.49 MB)
 Non-trainable params: 0 (0.00 B)
 Optimizer params: 1,303,582 (4.97 MB)
None
```

Evaluation

During the training part, we have validated the performance of our neural network model on images of single characters. Now it's time to test and evaluate CAPTCHAs from the beginning to the end. First, we will need to build the pipeline for CAPTCHA character prediction:

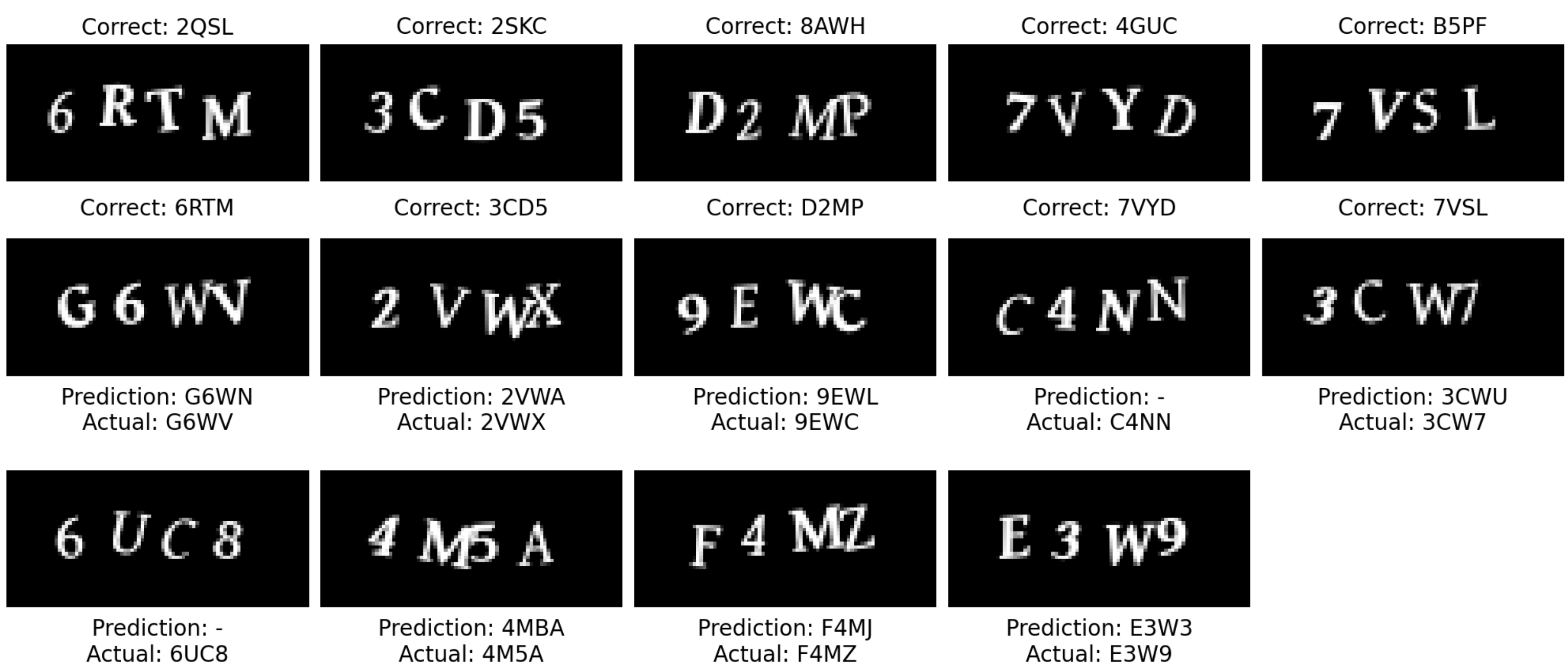
```
In [37]: # Load labels from file (so we can translate model predictions to actual letters)
         with open(LABELS_PATH, "rb") as f:
             lb = pickle.load(f)

         # Test our pipeline (and model) with the test set.
         # However, you'd want to replace this with some random CAPTCHAs in the real world.

         # Dummy character images
         DUMMY_CHAR_IMAGES = np.zeros((4, 20, 20, 1))

         # Indices of CAPTCHAs on which extractions failed
         extract_failed_indices = []
         # Extracted character images
         char_images_test = []

         # Extract character images and make features
         for i, captcha_image in enumerate(captcha_images_test):
             # Extract character images
             char_images = extract_chars(captcha_image)

             if char_images:
                 char_images_test.extend(char_images)
             # Use dummy character images as placeholder if extraction failed
             else:
                 extract_failed_indices.append(i)
                 char_images_test.extend(DUMMY_CHAR_IMAGES)

         # Make features for character images
         features_test = [make_feature(char_image) for char_image in char_images_test]
         # Scale raw pixel values into range [0, 1]
         features_test = np.array(features_test, dtype="float")/255

         # Make features for character images
         features_test = [make_feature(char_image) for char_image in char_images_test]
         # Scale raw pixel values into range [0, 1]
         features_test = np.array(features_test, dtype="float")/255
         # Predict labels with neural network
         preds_test = model.predict(features_test)
         # Convert one-hot encoded characters back
         preds_test = lb.inverse_transform(preds_test)

         # Group all 4 characters for the same CAPTCHA
         preds_test = ["".join(chars) for chars in group_every(preds_test, 4)]
         # Update result for CAPTCHAs on which extractions failed
         for i in extract_failed_indices:
             preds_test[i] = "-"
```

```
29/29 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step
```

Now, we can compute the accuracy of our pipeline, as well as taking a look at correct and incorrect CAPTCHA text predictions:

```
In [41]: # Number of CAPTCHAs to display
         N_DISPLAY_SAMPLES = 10

         # Number of test CAPTCHAs
         n_test = len(captcha_texts_test)
         # Number of correct predictions
         n_correct = 0

         # Indices of correct predictions
         correct_indices = []
         # Indices of incorrect predictions
         incorrect_indices = []

         for i, (pred_text, actual_text) in enumerate(zip(preds_test, captcha_texts_test)):
             if pred_text==actual_text:
                 ## [ TODO ]
                 # 1) Update number of correct predictions
                 n_correct += 1
                 # 2) Collect index of correct prediction
                 if len(correct_indices)<N_DISPLAY_SAMPLES:
                     correct_indices.append(i)
             else:
                 # 3) Collect index of incorrect prediction
                 if len(incorrect_indices)<N_DISPLAY_SAMPLES:
                     incorrect_indices.append(i)

         # Show number of total / correct predictions and accuracy
         print("# of test CAPTCHAs:", n_test)
         print("# correctly recognized:", n_correct)
         print("Accuracy:", n_correct/n_test, "\n")

         # Show all correct predictions
         print_images(
             [captcha_images_test[i] for i in correct_indices],
             texts=[f"Correct: {captcha_texts_test[i]}" for i in correct_indices],
             n_rows=2
         )

         # Show all incorrect predictions
         print_images(
             [captcha_images_test[i] for i in incorrect_indices],
             texts=[
                 f"Prediction: {preds_test[i]}\nActual: {captcha_texts_test[i]}" \
                 for i in incorrect_indices
             ],
             n_rows=2,
             fig_size=(20, 6),
             text_center=(0.5, -0.25)
         )
```

```
# of test CAPTCHAs: 228
# correctly recognized: 219
Accuracy: 0.9605263157894737

Figure(2000x500)
Figure(2000x600)
```


Correct: 2QSL Correct: 2SKC Correct: 8AWH Correct: 4GUC Correct: B5PF

Correct: 6RTM Correct: 3CD5 Correct: D2MP Correct: 7VYD Correct: 7VSL

Prediction: G6WN Actual: G6WV; Prediction: 2VWA Actual: 2VWX; Prediction: 9EWL Actual: 9EWC; Prediction: - Actual: C4NN; Prediction: 3CWU Actual: 3CW7

Prediction: - Actual: 6UC8; Prediction: 4MBA Actual: 4M5A; Prediction: F4MJ Actual: F4MZ; Prediction: E3W3 Actual: E3W9

Open-Ended Extension:

TensorFlow is one popular framework for implementing Deep Neural Networks, and another widely used framework is **PyTorch**. After you successfully complete this lab:

- Use your favorite language model and/or coding assistant (e.g., **GPT**, **Copilot**, **Gemini**, **Cursor**, etc.) to **convert your TensorFlow CNN implementation into PyTorch**.
- Repeat the same training + evaluation experiments in PyTorch.
- You should observe **similar results** (minor differences are expected due to randomness and implementation details).

✅ **Submission Requirement:**
Include this as a **separate notebook** in your lab-submission GitHub repository named:

```
Breaking-CAPTCHAS-Pytorch.ipynb
```

References

1. How to break a CAPTCHA system in 15 minutes with Machine Learning: https://medium.com/@ageitgey/how-to-break-a-captcha-system-in-15-minutes-with-machine-learning-dbebb035a710
2. CaptchaSolver Jupyter Notebook: https://github.com/BenjaminWegener/CaptchaSolver
3. Keras Tutorial: The Ultimate Beginner's Guide to Deep Learning in Python: https://elitedatascience.com/keras-tutorial-deep-learning-in-python
4. Keras API reference: https://keras.io/
5. Tensorflow API reference: https://www.tensorflow.org/api_docs